

Advanced (Habanero)

- Q1.** What is the factored form of $x^3 + y^3$?
- (A) $(x + y)(x^2 - xy + y^2)$
(B) $(x - y)(x^2 + xy + y^2)$
(C) $(x + y)(x^2 + xy - y^2)$
(D) $(x - y)(x^2 - xy - y^2)$
(E) $(x + y)(x^2 - xy - y^2)$
- Q2.** What is the factored form of $8x^3 - 27y^3$?
- (A) $(2x - 3y)(4x^2 + 6xy + 9y^2)$
(B) $(2x + 3y)(4x^2 - 6xy + 9y^2)$
(C) $(2x - 3y)(4x^2 - 6xy - 9y^2)$
(D) $(2x + 3y)(4x^2 + 6xy - 9y^2)$
(E) $(2x - 3y)(4x^2 + 6xy - 9y^2)$
- Q3.** Which of the following is a difference of cubes?
- (A) $x^3 + y^3$
(B) $x^3 - y^3$
(C) $x^2 + y^2$
(D) $x^2 - y^2$
(E) $x^4 - y^4$
- Q4.** What is the factored form of $64x^3 + 125y^3$?
- (A) $(4x + 5y)(16x^2 - 20xy + 25y^2)$
(B) $(4x - 5y)(16x^2 + 20xy + 25y^2)$
(C) $(4x + 5y)(16x^2 + 20xy - 25y^2)$
(D) $(4x - 5y)(16x^2 - 20xy - 25y^2)$
(E) $(4x + 5y)(16x^2 - 20xy - 25y^2)$
- Q5.** Which of the following expressions can be written as a sum of cubes?
- (A) $x^2 - y^2$
(B) $x^3 - y^3$
(C) $x^3 + y^3$
(D) $x^4 - y^4$
(E) $x^4 + y^4$
- Q6.** What is the factored form of $27x^3 - 8y^3$?
- (A) $(3x - 2y)(9x^2 + 6xy + 4y^2)$
(B) $(3x + 2y)(9x^2 - 6xy + 4y^2)$
(C) $(3x - 2y)(9x^2 - 6xy - 4y^2)$
(D) $(3x + 2y)(9x^2 + 6xy - 4y^2)$
(E) $(3x - 2y)(9x^2 + 6xy - 4y^2)$
- Q7.** What is the factored form of $x^3 - 64y^3$?
- (A) $(x - 4y)(x^2 + 4xy + 16y^2)$
(B) $(x + 4y)(x^2 - 4xy + 16y^2)$
(C) $(x - 4y)(x^2 - 4xy - 16y^2)$
(D) $(x + 4y)(x^2 + 4xy - 16y^2)$
(E) $(x - 4y)(x^2 + 4xy - 16y^2)$
- Q8.** Which of the following is NOT a sum or difference of cubes?
- (A) $x^3 + y^3$
(B) $x^3 - y^3$
(C) $x^2 + y^2$
(D) $x^2 - y^2$
(E) $x^4 - y^4$

Open Ended Questions (FRQ)

Q9. Factor the expression $8x^3 + 27y^3$ and show all steps.

Q10. Prove that $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$ using the distributive property.

Chef's Tips

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- Tip 1: Make sure to check the signs when factoring sums and differences of cubes.
- Tip 2: Use the formulas $(a + b)(a^2 - ab + b^2)$ and $(a - b)(a^2 + ab + b^2)$ to factor expressions.
- Tip 3: Pay close attention to the coefficients of the terms when factoring.